

TM Trading V9.2 Master Roadmap

From Research Chaos -> Clean Replay -> Verified Candidate -> Paper Trading

Purpose

This document is the control map for the TM Trading project. It prevents strategy work from exploding sideways across four repos, old branches, research notes, and temporary scripts.

1. Current Truth

Canonical repo: gitugu555/tm-trading-v92-core

This is the source of truth. It contains the safety helpers, dry-run manifest work, and the merged C_ExhaustionFade replay module.

Supporting repos:

- tm-trading-v92-core: canonical tested engine.
- tm-trading-research: sandbox only.
- tm-trading-v73-current: old benchmark / forensic comparison.
- tm-trading-v555: legacy memory and data lineage.

Rule: no new canonical strategy logic goes into v555, v73-current, or research. Everything real graduates into v92-core with tests.

2. Strategy Status

A_Breakout: RESEARCH_ONLY. It looked promising in raw surface tests, but weakened after de-overlap and effective sample checks.

B_Absorption: DATA_BLOCKED. It needs real OFI/L2 coverage. Do not fake it with proxy data.

C_ExhaustionFade: KEEP. This is the current main candidate.

Current C definition:

- bar_size: 750 BTC
- regime: EXHAUSTED
- side: long-only
- entry: next-bar open
- exit: h36 time exit
- position rule: one position at a time
- cost model: explicit flat round-trip bps

3. What The Session Achieved

The project moved from "we found something interesting in research output" to "we have a clean, tested, merged replay harness for the first surviving candidate."

Merged into master:

- replays/__init__.py
- replays/c_exhaustion_replay.py
- scripts/run_c_exhaustion_replay.py
- tests/conftest.py
- tests/test_c_exhaustion_replay.py
- tests/test_run_c_exhaustion_replay_cli.py

The CLI blocks output under:

- data/hft/tier2
- data/hft
- data/raw

4. Immediate Next Mission: Reproduce Phase 1I

Run the merged CLI against the validated 750 BTC parquet shards.

Command:

```
python scripts/run_c_exhaustion_replay.py \  
--bar-dir /path/to/750btc/shards \  
--bar-size 750 \  
--horizon 36 \  
--cost-bps 12 \  
--output-dir reports/c_exhaustion_replay_real_750_h36_cost12 \  
--write-json \  
--write-csv \  
--write-summary
```

Compare against Phase 1I memory:

- trade_count: about 221
- net_expectancy_bps: about 72.23 at cost12
- profit_factor: about 2.19
- sharpe: about 1.63
- worst_year: 2025

Decision:

- Close match: replay harness confirmed; commit reproduction report.

- Mild mismatch: debug classifier/cost/calendar differences.
- Big mismatch: stop downstream work and debug before trusting the signal.

5. Reproduction Report Rule

Commit small summaries only:

- reports/c_exhaustion_replay_real_750_h36_cost12_summary.md
- reports/c_exhaustion_replay_real_750_h36_cost12_report.json

Do not commit large CSV logs, parquet shards, raw data, or cache files.

6. After Reproduction: Small Robustness Grid

Do not launch a huge parameter sweep. Test only the local neighborhood around C.

Grid:

- bar_size: 500, 750, 1000
- horizon: 24, 36, 48
- cost_bps: 5, 8, 12, 20
- side: long-only only
- entry: next-bar open only
- exit: time exit only

Goal: prove C is not a single lucky 750/h36 island.

Verdict:

- STRONG: 750/h36 reproduces and neighboring configs remain positive at cost12.
- CAUTION: only one island works.
- FAIL: reproduction or neighbors collapse.

7. C Candidate Graduation Ladder

Gate 1 - Code safety: PASSED.

Gate 2 - Real replay reproduction: NEXT.

Gate 3 - Local robustness: NOT STARTED.

Gate 4 - Paper-trading simulator: NOT STARTED.

Gate 5 - Execution adapter: DO NOT START YET.

8. Do Not Build Yet

Avoid these until C reproduction and robustness are done:

- live bot
- Jarvis agent
- Telegram execution
- Nautilus live adapter
- multi-strategy allocator
- Kelly sizing
- portfolio optimizer
- AI decision layer

9. Data Foundation Direction

C can be tested with existing 750 BTC bars. B cannot be tested honestly without L2/OFI.

L2/OFI tasks:

- 1. Inventory available L2 sources.**
- 2. Confirm exchange, symbol, depth, date range, format, cost.**
- 3. Build raw L2 schema.**
- 4. Build OFI feature cache.**
- 5. Add coverage-preserving joins.**
- 6. Test B_Absorption only when coverage is real.**

Rule: no OFI coverage = no B signal.

10. Repo Hygiene Rules

Every serious change gets a branch.

A PR must have: single purpose, small scope, tests, no generated bytecode, no data files, no unrelated drift.

Merge only when: pytest passes, changed files are expected, no production write risk, no lookahead risk, no duplicated strategy truth without comment.

11. Indicator Research Appendix

Claude is right to warn against adding indicator combos before reproduction. Indicator research is allowed only after Gate 2, and ideally after Gate 3. The goal is not to bolt on RSI/MACD until winrate rises. The goal is to test whether a small number of conceptually aligned confirmations improve robustness without overfit.

Allowed later research families:

A. Exhaustion confirmation:

- RSI divergence near rolling low
- ATR/ADR stretch normalization
- declining volume after extreme push
- candle rejection / close-quality confirmation

B. Order-flow confirmation:

- CVD divergence
- delta exhaustion
- negative delta into lower low with failed continuation
- OFI only when real coverage exists

C. Exit refinement:

- volatility-scaled time exit
- ATR emergency stop
- MFE/MAE-informed protective stop
- no TP/SL curve fitting; Phase 1H already showed TP/SL can clip edge

D. Regime boundary improvement:

- improve EXHAUSTED label, not random confluence stacking
- compare central regime classifier against replay fallback parity

Anti-overfit rules:

- no huge combinatorial sweeps
- no optimizing thresholds before reproduction
- require walk-forward, de-overlap, yearly split, and cost stress
- improvement must hold at cost12 and preferably cost20
- 2025 cannot be hidden by an overfit no-trade filter

12. Codex Prompt: Reproduce C on Real Data

Use the merged script. Do not modify source code first. Do not write to production cache paths. Extract `trade_count`, `net_expectancy_bps`, `win_rate`, `profit_factor`, `sharpe`, `max_drawdown_pct`, `positive_year_count`, `worst_year`, and `yearly_rows`. Compare with Phase 1I. Produce a PASS / INVESTIGATE / FAIL verdict. Do not commit until reviewed.

13. Final Operating Principle

Do not build the whole trading firm at once. Confirm one edge. Reproduce it. Stress it. Paper trade it. Then add the next layer.

Right now the whole project depends on one question: does the merged C replay reproduce Phase 1I on real 750 BTC shards?